1 Claude version

Detecting the memory parameter of a heavy-tailed MA process via TDA

Setting. Let $(\varepsilon_t)_{t\in\mathbb{Z}}$ be a sequence of independent identically distributed heavy-tailed innovations with tail index $\alpha > 0$, in the sense that

$$\mathbb{P}(|\varepsilon_t| > x) \;\sim\; L(x)\, x^{-\alpha}, \qquad x \to \infty,$$

for some slowly varying function L. Consider a moving-average process of unknown order $m \in \mathbb{N}$,

$$X_t \;=\; \sum_{j=0}^{m} \theta_j\, \varepsilon_{t-j}, \qquad t = 1, \dots, T, \tag{1}$$

with deterministic coefficients $\theta_0 = 1$ and $\theta_m \neq 0$. The aim is to estimate the memory parameter m from a single realisation of length T using topological data analysis.

Sliding-window (delay) embeddings and the dimension drop. For an integer dimension $k \geq 1$, define the *k-dimensional sliding-window embedding* of the time series by

$$Y_t^{(k)} \;:=\; (X_t, X_{t+1}, \dots, X_{t+k-1})^\top \;\in\; \mathbb{R}^k, \qquad t = 1, \dots, T-k+1, \tag{2}$$

and let

$$\mathcal{P}_k \;:=\; \left\{\, Y_t^{(k)} \;\middle|\; t = 1, \dots, T-k+1 \,\right\} \;\subset\; \mathbb{R}^k \tag{3}$$

be the resulting overlapping-window point cloud. The cloud $\mathcal{P}_k$ has cardinality $n_k := T-k+1$ and inherits an intrinsic dimensionality from the recursive structure of (1).

The key observation is that consecutive embedded points $Y_t^{(k)}$ and $Y_{t+1}^{(k)}$ share $k-1$ coordinates, and each $Y_t^{(k)}$ is a linear function of the $k+m$ innovations $\varepsilon_{t-m}, \varepsilon_{t-m+1}, \dots, \varepsilon_{t+k-1}$. Since the embedding map $(\varepsilon_{t-m}, \dots, \varepsilon_{t+k-1}) \mapsto Y_t^{(k)}$ factors through an $\mathbb{R}^{k+m}$-to-$\mathbb{R}^k$ linear map of rank exactly $\min(k, k+m) = k$ when $k \leq m$, the point cloud $\mathcal{P}_k$ spans a subspace of full dimension k for all $k \leq m+1$. For $k > m+1$, however, every consecutive pair $Y_t^{(k)}, Y_{t+1}^{(k)}$ satisfies the linear constraint

$$\sum_{j=0}^{m} \theta_j \big(Y_{t+1}^{(k)} - L\, Y_t^{(k)}\big)_? \;=\; 0$$

inherited from the MA recursion, where L is the appropriate shift; the upshot is that $\mathcal{P}_k$ concentrates near an $(m+1)$-dimensional affine subspace of $\mathbb{R}^k$. Consequently:

- For $k \leq m+1$: $\mathcal{P}_k$ is genuinely k-dimensional.
- For $k = m+1$: $\mathcal{P}_k$ first reaches its intrinsic dimension and exhibits "critical" geometric structure.
- For $k > m+1$: $\mathcal{P}_k$ is approximately $(m+1)$-dimensional in the ambient $\mathbb{R}^k$, with deviation governed by the heavy-tailed fluctuations of the innovations.

The transition $k = m \to k = m+1$ is the topologically critical step at which the intrinsic dimensionality of $\mathcal{P}_k$ stops growing. Detecting this transition is the goal of the construction below.

Construction of the simplicial filtration. For each embedding dimension $k = 1, 2, \ldots, K$ (where K is an a priori upper bound for the order of the MA process, $K \gg m$), build the *Vietoris–Rips filtration* of the point cloud $\mathcal{P}_k$:

$$\mathrm{VR}_r(\mathcal{P}_k) \;:=\; \big\{\, \sigma \subseteq \mathcal{P}_k \;\big|\; \mathrm{diam}(\sigma) \le r \,\big\}, \qquad r > 0. \tag{4}$$

For a fixed reference scale $r_k > 0$, set

$$K_k \;:=\; \mathrm{VR}_{r_k}(\mathcal{P}_k),$$

where r_k is chosen as a quantile (for instance the median) of the pairwise distance distribution within $\mathcal{P}_k$, ensuring scale-invariance under the heavy-tailed distribution of X_t.

This produces a sequence of simplicial complexes $K_1, K_2, \ldots, K_K$, indexed by the embedding dimension k. Since the point clouds $\mathcal{P}_k$ live in different ambient spaces, the K_k are not literally a filtration in the inclusion sense, but they form a *family* of complexes parameterised by k, and a complexity functional applied to each K_k produces a sequence of values whose behaviour as a function of k encodes the intrinsic dimensionality.

Topological complexity functional. The intrinsic-dimension transition is most directly captured by a functional that responds to the *ambient dimension actually filled* by the point cloud. We propose using the highest-dimensional non-trivial Betti number, or equivalently a weighted Betti-number sum:

$$f(K_k) \;:=\; \sum_{j=0}^{\dim K_k} w_j \, \beta_j(K_k; \mathbb{F}_2), \tag{5}$$

where $\beta_j(K_k; \mathbb{F}_2) = \dim_{\mathbb{F}_2} H_j(K_k; \mathbb{F}_2)$ is the j-th Betti number with $\mathbb{F}_2$-coefficients, and the weights $w_j > 0$ emphasise higher-dimensional features. A simple and well-motivated choice is $w_j = j$, so that

$$f(K_k) \;=\; \sum_{j=1}^{\dim K_k} j \cdot \beta_j(K_k; \mathbb{F}_2), \tag{6}$$

which is sensitive to the existence of higher-dimensional voids in K_k and is large when $\mathcal{P}_k$ is genuinely high-dimensional, but substantially smaller when $\mathcal{P}_k$ concentrates on a low-dimensional subspace. Alternative choices include the LS category $\mathrm{cat}(K_k)$, the cup length $\mathrm{cup}_{\mathbb{F}_2}(K_k)$, or magnitude $\mathcal{M}_{\mathcal{P}_k}(t_k)$ at a chosen scale t_k.

Heuristic for the parameter m. Under the dimension-drop hypothesis, the sequence

$$\big(f(K_1), f(K_2), \ldots, f(K_K)\big)$$

should grow approximately linearly (or polynomially) in k for $k \le m + 1$, and *plateau* or grow much more slowly for $k > m + 1$, reflecting the fact that increasing the embedding dimension beyond $m + 1$ no longer reveals new topological structure. The transition index — the smallest k at which the growth of $f(K_k)$ becomes sublinear — is a candidate estimator of m.

Scale similarity matrix and Perron eigenvector. To formalise this in the framework of the present project, we set $a_k := f(K_k)$ for $k = 1, \ldots, K$, and (since the sequence is monotone non-decreasing under the dimension hypothesis) define the scale similarity matrix $A \in \mathbb{R}^{K \times K}$ with entries

$$A_{ij} \;=\; a_{\min(i,j)} \;=\; f(K_{\min(i,j)}), \qquad 1 \le i, j \le K. \tag{7}$$

By construction, A is symmetric, positive, and irreducible, and admits a unique positive Perron eigenvector $v_K = (v_{K,1}, \ldots, v_{K,K})^\top$ associated with its spectral radius λ_K (Proposition **??**

above). By the comonotonicity of v_K and $(a_k)_{k=1}^K$, the entries of v_K inherit the same monotonicity profile as $f(K_k)$.

The estimate of the memory parameter is then

$$\widehat{m} \;:=\; \operatorname*{arg\,max}_{k\in\{1,\dots,K\}} v_{K,k} \;-\; 1, \tag{8}$$

i.e., one less than the embedding dimension at which the Perron eigenvector attains its maximum. Geometrically, $\widehat{m}$ is the largest k for which the topological complexity continues to grow with the embedding dimension; subtracting one accounts for the fact that the dimension drop occurs at $k = m + 1$.

Robustness considerations for heavy-tailed innovations. The use of $\mathbb{F}_2$-Betti numbers in (5) is deliberate: integer-coefficient Betti numbers are sensitive to torsion that may arise from heavy-tailed outliers, whereas the field $\mathbb{F}_2$ gives a robust dimension count. The choice of reference scale r_k as a quantile of the pairwise-distance distribution (rather than as an absolute threshold) further provides scale-invariance under the heavy-tailed marginal distribution of X_t, ensuring that the procedure is asymptotically distribution-free in the tail behaviour of the innovations. Theoretical justification of the consistency of $\widehat{m}$ under (1) for $\alpha > 2$ (finite variance) is a direct target of WP 3 of the present project; the heavy-tailed regime $0 < \alpha \leq 2$ requires more delicate analysis through the tail-dependence machinery developed in our prior work on max-stable processes.

2 ChatGPT version

$$X_t = \sum_{j=0}^{m-1} a_j Z_{t-j}, \qquad t \in \mathbb{Z},$$

where the innovations Z_t are assumed to be heavy-tailed. The unknown integer m is the memory parameter. We observe a finite sample

$$X_1, \dots, X_N.$$

For every candidate tuple length $d = 1, \dots, d_{\max}$, divide the time series into non-overlapping blocks of length d:

$$Y_k^{(d)} = \big(X_{(k-1)d+1}, X_{(k-1)d+2}, \dots, X_{kd}\big) \in \mathbb{R}^d, \qquad k = 1, \dots, n_d,$$

where

$$n_d = \left\lfloor \frac{N}{d} \right\rfloor.$$

Since the innovations are heavy-tailed, the radial magnitude of the points can dominate the geometry. Therefore we pass to directions. Define

$$U_k^{(d)} = \frac{Y_k^{(d)}}{\|Y_k^{(d)}\|_2} \in \mathbb{S}^{d-1},$$

whenever $Y_k^{(d)} \neq 0$. The point cloud at tuple length d is

$$\mathcal{P}_d = \left\{ U_1^{(d)}, \dots, U_{n_d}^{(d)} \right\} \subset \mathbb{S}^{d-1}.$$

Because a linear dependence is invariant under sign, one may use the projective metric

$$\rho_d(u, v) = \min\{\|u - v\|_2, \|u + v\|_2\}, \qquad u, v \in \mathbb{S}^{d-1}.$$

For every scale parameter $\varepsilon > 0$, define the Vietoris–Rips complex

$$K_d(\varepsilon) = \mathrm{VR}(\mathcal{P}_d, \rho_d, \varepsilon).$$

Thus a finite subset

$$\sigma = \{u_{i_0}, \ldots, u_{i_q}\} \subset \mathcal{P}_d$$

is a simplex of $K_d(\varepsilon)$ if

$$\rho_d(u_{i_r}, u_{i_s}) \leq \varepsilon \qquad \text{for all } 0 \leq r, s \leq q.$$

The family

$$\mathcal{K} = \{K_d(\varepsilon) : d = 1, \ldots, d_{\max},\ \varepsilon \geq 0\}$$

is the desired filtration of simplicial complexes, with one filtration for each tuple dimension d.

For each vertex $u_i \in \mathcal{P}_d$, let

$$\mathrm{St}_{K_d(\varepsilon)}(u_i)$$

denote the closed star of u_i in the complex $K_d(\varepsilon)$. Let

$$\mathcal{N}_i^{(d)}(\varepsilon) = \left\{u_j \in \mathcal{P}_d : u_j \in \mathrm{St}_{K_d(\varepsilon)}(u_i)\right\}$$

be the local simplicial neighbourhood of u_i at scale ε.

Define the local covariance matrix

$$C_i^{(d)}(\varepsilon) = \frac{1}{|\mathcal{N}_i^{(d)}(\varepsilon)|} \sum_{u_j \in \mathcal{N}_i^{(d)}(\varepsilon)} \left(u_j - \bar{u}_i^{(d)}(\varepsilon)\right)\left(u_j - \bar{u}_i^{(d)}(\varepsilon)\right)^{\top},$$

where

$$\bar{u}_i^{(d)}(\varepsilon) = \frac{1}{|\mathcal{N}_i^{(d)}(\varepsilon)|} \sum_{u_j \in \mathcal{N}_i^{(d)}(\varepsilon)} u_j.$$

Let

$$\lambda_1\left(C_i^{(d)}(\varepsilon)\right) \geq \cdots \geq \lambda_d\left(C_i^{(d)}(\varepsilon)\right) \geq 0$$

be the eigenvalues of $C_i^{(d)}(\varepsilon)$. If the local point cloud lies near a hyperplane in $\mathbb{R}^d$, then the smallest eigenvalue should be small. Therefore define the local hyperplane-dependence score

$$H_i^{(d)}(\varepsilon) = \exp\left(-\frac{\lambda_d\left(C_i^{(d)}(\varepsilon)\right)}{\lambda_1\left(C_i^{(d)}(\varepsilon)\right) + \cdots + \lambda_d\left(C_i^{(d)}(\varepsilon)\right) + \delta} \cdot \frac{1}{\gamma}\right),$$

where $\delta > 0$ is a numerical regularization parameter and $\gamma > 0$ controls the sensitivity. The quantity $H_i^{(d)}(\varepsilon)$ is close to 1 when the local neighbourhood is nearly contained in a hyperplane, and is smaller when the neighbourhood is genuinely d-dimensional.

Now define the persistent simplicial hyperplane functional

$$\Phi_d(\varepsilon) = \frac{\sum_{i=1}^{n_d} w_i^{(d)}(\varepsilon) H_i^{(d)}(\varepsilon)}{\sum_{i=1}^{n_d} w_i^{(d)}(\varepsilon)},$$

where the weights may be chosen as

$$w_i^{(d)}(\varepsilon) = \#\mathcal{N}_i^{(d)}(\varepsilon),$$

or, more topologically,

$$w_i^{(d)}(\varepsilon) = \#\left\{\sigma \in K_d(\varepsilon) : u_i \in \sigma\right\}.$$

Thus $\Phi_d(\varepsilon)$ is large when the simplicial neighbourhoods of the point cloud $\mathcal{P}_d$ are locally close to hyperplanes. This is the proposed TDA functional.

To obtain a scale-independent signature for the tuple dimension d, integrate over the filtration parameter:

$$A_d = \int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \Phi_d(\varepsilon)\, \omega_d(\varepsilon)\, d\varepsilon,$$

where $\omega_d(\varepsilon)$ is a persistence weight. A natural choice is

$$\omega_d(\varepsilon) = \frac{\beta_0\big(K_d(\varepsilon)\big)^{-1}}{\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \beta_0\big(K_d(s)\big)^{-1}\, ds},$$

so that scales where the complex has begun to connect, but has not yet become completely saturated, receive higher weight.

Equivalently, define the full functional profile

$$f_d(\varepsilon) = \Phi_d(\varepsilon), \qquad \varepsilon \in [\varepsilon_{\min}, \varepsilon_{\max}].$$

The scale similarity matrix between tuple dimensions is then defined by

$$S_{d\ell} = \exp\left(-\frac{\|f_d - f_\ell\|^2_{L^2(\omega)}}{\tau^2}\right), \qquad 1 \le d, \ell \le d_{\max},$$

where

$$\|f_d - f_\ell\|^2_{L^2(\omega)} = \int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \big(f_d(\varepsilon) - f_\ell(\varepsilon)\big)^2\, \omega(\varepsilon)\, d\varepsilon.$$

Let $v \in \mathbb{R}^{d_{\max}}$ be the Perron–Frobenius eigenvector of S:

$$Sv = \lambda_{\max}(S)v, \qquad v_d \ge 0, \qquad \|v\|_2 = 1.$$

The estimated memory parameter is then

$$\widehat{m} = \arg \max_{1 \le d \le d_{\max}} v_d.$$

The interpretation is that the true memory scale m should be the tuple dimension whose persistent hyperplane-dependence profile is most structurally central among the profiles $\{f_d\}_{d=1}^{d_{\max}}$. In particular, when $d < m$, the tuple cloud should not yet exhibit the full linear dependence structure induced by the moving-average memory. Around $d = m$, the local simplicial neighbourhoods should become maximally close to a lower-dimensional linear object. For $d > m$, the same dependence structure is embedded into larger-dimensional spaces and the signal becomes less concentrated. Hence the coordinate v_d of the principal eigenvector of the similarity matrix is expected to peak at $d = m$.

A sharper version uses the discrete transition profile

$$\Delta_d = \|f_{d+1} - f_d\|_{L^2(\omega)}, \qquad d = 1, \dots, d_{\max} - 1.$$

Then one may form the transition-weighted similarity matrix

$$\widetilde{S}_{d\ell} = S_{d\ell} \exp\left(-\alpha |\Delta_d - \Delta_\ell|^2\right),$$

and estimate

$$\widehat{m} = \arg\max_d \widetilde{v}_d,$$

where

$$\widetilde{S}\widetilde{v} = \lambda_{\max}(\widetilde{S})\widetilde{v}.$$

In practice, the recommended functional is therefore

$$\boxed{\Phi_d(\varepsilon) = \frac{\sum_{i=1}^{n_d} w_i^{(d)}(\varepsilon) \exp\left(-\frac{\lambda_d\left(C_i^{(d)}(\varepsilon)\right)}{\operatorname{tr} C_i^{(d)}(\varepsilon)+\delta} \cdot \frac{1}{\gamma} \right)}{\sum_{i=1}^{n_d} w_i^{(d)}(\varepsilon)}}$$

together with the similarity matrix

$$\boxed{S_{d\ell} = \exp\left(-\frac{\int \left(\Phi_d(\varepsilon) - \Phi_\ell(\varepsilon)\right)^2 \omega(\varepsilon)\, d\varepsilon}{\tau^2} \right)}$$

and the estimator

$$\boxed{\widehat{m} = \arg\max_d v_d, \qquad Sv = \lambda_{\max}(S)v.}$$

This is not ordinary persistent homology of holes; it is a persistent simplicial flatness statistic. That is the right kind of functional here, because the signal you want is not primarily the birth and death of cycles, but the emergence of approximate linear dependence in the d-tuple point clouds.